Why is it a lower bound?

$$\log p_\theta(x) = \mathrm{ELBO}(x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x))$$

The gap is nonnegative.

$$\mathrm{ELBO}(x) = \mathbb{E}_q[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q \| p)$$

How does diagonal-Gaussian KL scale to d dimensions?

$$D_{\mathrm{KL}}\left(\mathcal{N}(\mu, \mathrm{diag}(\sigma^2)) \,\|\, \mathcal{N}(0, I)\right) = \frac{1}{2}\sum_{j=1}^{d}\left(\mu_j^2 + \sigma_j^2 - \log\sigma_j^2 - 1\right)$$

With logvar: sum −½(1 + logvar − μ² − exp(logvar)).

> Sum over latent dimensions. Average over the batch.

Classical minimax versus practical generator loss

$$\min_G \max_D \; \mathbb{E}_{\text{data}} \log D(x) + \mathbb{E}_z \log(1 - D(G(z)))$$

Classical saturating G objective: minimize the second term.

$$L_G^{\text{non-saturating}} = -\mathbb{E}_z \log D(G(z))$$

> The main lecture uses the common non-saturating objective.

Backup · optional

What exactly was measured?

Model	Test recon. nats	Test KL nats
ae2	133.20	0.00
vae2	135.39	6.34
vae16	69.96	22.02

Binary threshold >127 · Adam 0.001 · batch 256 · 15 epochs · CPU · seed 667

Decoder: d → 128 → 256 → 784 logits. ReLU hidden layers.

Encoder: 784 → 256 → 128 → μ / logvar. β = 1.

Sources and reproducibility

Explanations and course material

- Lilian Weng: From Autoencoder to Beta-VAE
- Lilian Weng: From GAN to WGAN
- Jaan Altosaar: What is a variational autoencoder?
- Nipun Batra: Variational Inference, Part 2
- Nipun Batra: Variational Autoencoders
- Nipun Batra: PML VAE notebook
- MIT 6.S191: Deep Generative Modeling

Papers and worked implementations

- Kingma & Welling: Auto-Encoding Variational Bayes
- Goodfellow et al.: Generative Adversarial Networks
- Rezende & Mohamed: Normalizing Flows
- TensorFlow: Convolutional Variational Autoencoder
- PyTorch: Pathwise derivative and rsample

Complete resource index: resources.html in the lecture bundle.